

6 Fourier Coefficients

Take a square wave from $-T_0$ to T_0

We can Fourier decompose this with

$$a_x = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$$

$$\text{where } \omega_0 = \frac{2\pi}{T_0}$$

$$A_k = \frac{1}{T_0} \left[\int_{-\frac{T_0}{2}}^0 (-1) e^{-jk\omega_0 t} dt + \int_0^{\frac{T_0}{2}} (1) e^{-jk\omega_0 t} dt \right]$$

$$\left[\frac{1}{T_0} \frac{-1}{-jk\omega_0} \underbrace{e^{-jk\omega_0 t}}_{-\frac{T_0}{2}}^0 + \frac{1}{-jk\omega_0} \underbrace{e^{-jk\omega_0 t}}_0^{\frac{T_0}{2}} \right]$$

$$\left[\frac{1}{T_0} \left(\frac{1 - e^{-jk\omega_0 \frac{T_0}{2}}}{-jk\omega_0} - \frac{1 - e^{-jk\omega_0 \frac{T_0}{2}}}{-jk\omega_0} \right) \right]$$

For which the e to the stuff collapse to

$$(1 - e^{jk\pi}) \text{ and } (e^{-jk\pi} - 1)$$

We can fully unpack these to get

$$a_k = \frac{1}{jk\pi} (1 - \cos(k\pi))$$

$$= \frac{1}{jk\pi} [1 - (-1)^k]$$

Properties of the Fourier series

Continuous time

Linearity

Coefficients are scaled with the function

Time shift: If we move a function by some amount, we multiply each coefficient by the complex exponential

$$a(t) \rightarrow a_k$$

$$b(t) \rightarrow a(t - t_0)$$

$$b_k = e^{-jk\omega_0 t_0} a_k$$

Discrete time

$$x[n] = x[n + N], N \text{ is the fundamental period}$$

$$\hat{\omega}_0 = \frac{2\pi}{N} \text{ is the fundamental frequency}$$

This is conceptually similar but with a big wrinkle.

$$\phi_k[n] = e^{jk\hat{\omega}_0 n}, k = 0, \pm 1, \pm 2, \dots$$

Instead of summing over all infinite frequencies, we have to limit ourselves. This is because the k's are no longer unique:

$$e^{jk\hat{\omega}_0 n} = e^{jk\hat{\omega}_0 n(k+rN)}$$

where r is an integer and N is the fundamental period

$$\begin{aligned} &= e^{jk\omega_0 n} e^{jrN \frac{2\pi}{N} n} \\ &= e^{jk\omega_0 n} \end{aligned}$$

since

$$e^{2\pi rjn} = \cos(2\pi \underbrace{rm}_{\text{integer}}) + j \sin(2\pi rm) = 1$$

If, for example, our signal is periodic over $n = 4$, then we only have k from -4 to 4 , otherwise we would repeat.

Wild thing: We can now use any range of k for our basis functions, as long as there are all N versions.

Can we really reproduce a signal with such a tight limit of basis functions?

Lets take a vanilla range

$$x[n] = \sum_{k=0}^{N-1} a_k e^{jk\omega_0 n}$$

Lets multiply each side by $e^{-jr\omega_0 n}$.

We now sum over n from 0 to $N - 1$.

We get

$$\sum_{n=0}^{N-1} x[n] e^{-jr\omega_0 n} = \sum_{n=0}^{N-1} \sum_{k=0}^{N-1} a_k e^{jk\omega_0 n} e^{-jr\omega_0 n}$$

We can rewrite the right hand side:

$$\sum_{k=0}^{N-1} a_k \sum_{n=0}^{N-1} e^{j(k-r)\omega_0 n}$$

There is a lemma that we can use

$$\sum_{n=0}^{N-1} e^{jm\omega_0 n} = \begin{cases} N & m = 0, \pm N, \pm 2N \\ 0 & \text{otherwise} \end{cases}$$

For our cases, this is

$$\sum_{n=0}^{N-1} e^{j(k-r)\omega_0 n} = \begin{cases} N & k = r \\ 0 & k \neq r \end{cases}$$

Note that this is because we chose to start at 0 instead of any of the arbitrary set of consecutive numbers of length N that we could have used. We would have a different constraint of $k - r = n$ (or $k - r$) is an integer multiple of n . We don't span the range between each of the solutions $0, \pm N, \pm 2C$ etc. So we only need one - and we chose the simplest, which was zero.

$$a_r = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jr\omega_0 n}$$